# Mukul Rayana

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## TECHNICAL SKILLS

**Languages:** Python, C++, C#, SQL

**Software Engineering:** Data structures & algorithms, object-oriented design, REST APIs, Git, CI/CD, debugging & testing

**AI/ML:** PyTorch, Hugging Face Transformers, LangGraph/LangChain, QLoRA/PEFT, RAG, PPO/Ray RLlib, scikit-learn

**Databases & Search:** PostgreSQL, MongoDB, SQLite, pgvector, Pinecone, FAISS, ChromaDB

**Cloud & Deployment:** Docker, FastAPI, GitHub Actions, MLflow, AWS (EC2, S3, EKS), Azure Container Apps, Kubernetes

## EDUCATION

**University of Maryland, College Park** Aug 2024 - May 2026
*M.S., Applied Machine Learning* College Park, MD

Coursework: Algorithms & Data Structures, Computing Systems for ML, Cloud Computing, Optimization, NLP, Computer Vision.

**SRM Easwari Engineering College** Dec 2020 - Apr 2024
*B.Tech., Artificial Intelligence & Data Science* Chennai, India

Coursework: Programming & Data Structures, Object-Oriented Programming, DBMS (SQL/NoSQL), Deep Learning.

**Certification:** Data Structures & Algorithms Specialization, UC San Diego.

## EXPERIENCE

**BeeBox Studios Pvt Ltd | IIT Madras Research Park** Apr 2023 - Apr 2024
*AI Research Intern* Chennai, India

- Built a BERT question-answering system for in-headset lookup over a 100-page XR corpus, achieving **83.2% F1 and 71.8% exact match**. Reduced P95 inference latency from **185 ms to 62 ms** using ONNX and INT8 quantization on GPU-constrained hardware.
- Integrated the Python model with Unity/C# over TCP as the sole ML engineer working with UI/UX and WebRTC teams. Configured AWS EC2 and EKS and deployed the application using Docker and GitHub Actions CI/CD, supporting the team's **six production releases without service interruption**.

## PROJECTS

**PharmaChain: Pharmaceutical Cold-Chain Incident Response** Apr 2026

- Won **1st place ($4,000)** at the UMD Smith Agentic AI Challenge with a five-person team. Helped design an eight-agent LangGraph prototype for pharmaceutical shipment incidents, with human approval before irreversible actions.
- Built Route and Triage agents with REST endpoints for route selection and risk-based shipment ranking. Implemented **four-provider LLM failover** with a rule-based fallback and fixed an orchestration bug, eliminating **one redundant LLM call per incident**.

**Irminsul: LLM Fine-Tuning and RAG Serving** Nov 2025 - Jan 2026

- Fine-tuned Llama 3.1 8B with QLoRA on **52K instruction examples** and compared three MLflow experiments. Built a Docker/FastAPI RAG application; its Hugging Face Spaces demo uses a separate Groq-hosted Llama 3.3 70B backend.
- Automated weekly updates to an **840-document corpus and 6,876-vector Pinecone index** with GitHub Actions and incremental scraping. Implemented typed API responses, a health endpoint, prompt-injection checks and domain-relevance filtering.

**RECON: Multi-Agent Literature Research** Oct 2025 - Jan 2026

- Built a four-agent LangGraph pipeline for ML literature research using Semantic Scholar's REST API, recency-weighted retrieval and targeted search retries. Returned findings with claim-level confidence, source attribution and SQLite session memory.
- Evaluated five configurations on a **130-question benchmark** grounded in survey abstracts. Linear recency weighting flagged stale evidence on **52% of the 50 superseded-claim questions**, versus **42% without recency decay**.

**EmpathRAG: Mental Health RAG Research Prototype** Feb 2026 - Apr 2026

- Built a five-stage pipeline with a DeBERTa-v3 NLI guardrail before generation. Fine-tuned on **185,659 training pairs**; achieved **96.3% crisis recall and 79.5% precision** on a held-out set of **23,208 examples**.
- Implemented emotion-conditioned retrieval over **1.67M FAISS vectors** and Captum Integrated Gradients for token-level safety explanations. Caught **17 of 20 adversarial crisis probes**, versus **9 of 20** for a keyword filter.

## PUBLICATION

**Voice-Driven Panoramic Imagery: Real-Time Generative AI for Immersive Experiences.**

Co-author, IEEE IDCIoT 2024. Built speech-to-360-degree imagery with Whisper, GPT-Neo and Stable Diffusion; **2.3x throughput** using attention slicing and xFormers optimizations.